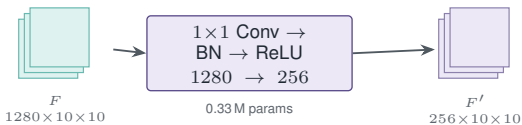
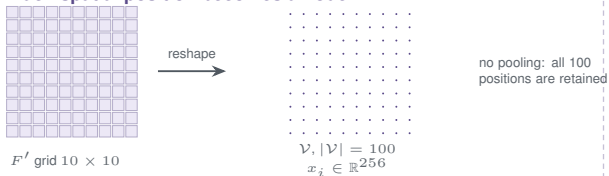


(A) Channel reduction

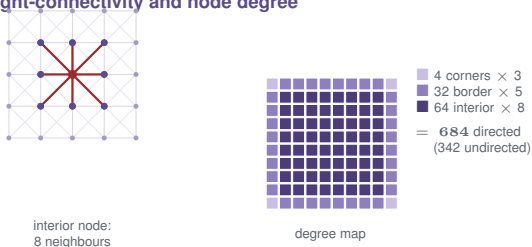


A single graph layer on the raw 1,280 channels would need a 1280×1280 weight matrix ($\approx 1.64 \text{ M}$); after projection each layer is 256×256 (65.5 K).

(B) Each spatial position becomes a node



(C) Eight-connectivity and node degree



(D) Self-loops and symmetric normalization

$$\tilde{A} = A + \mathbf{I}_N, \quad \tilde{D} = \text{diag}(\sum_j \tilde{A}_{ij}), \quad \hat{A} = \tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2}$$

Because the degree varies with position, $\hat{A}$ weights border and corner patches differently from interior ones—a property a fixed convolution kernel does not reproduce (see Discussion).